FEDERAL SPECIFICATION

CARBON DISULFIDE, TECHNICAL GRADE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 Grade.—This specification covers one grade of carbon disulfide for technical use.

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 Specifications. — The following Federal Specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

VV–L–791—Lubricants, Liquid Fuels, and Related Products; Methods of Inspection, Sampling and Testing.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C., or to the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Seattle. Prices may be obtained from the Superintendent of Documents or from the GSA Regional Offices.)

2.2 Other publication. — The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

Code of Federal Regulations, Title 49, Transportation, Chapter 1, Interstate Commerce Commission, Parts 71–78: Explosives and Other Dangerous Articles—49 CFR 71–78.

(Copies of 49 CFR 71–78 are obtainable from the Superintendent of Documents, Government Printing Office, Washington 25, D. C., price $3.75.)

2.3 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Physical and chemical requirements.—Carbon disulfide, technical grade, shall meet the requirements given in table I, when tested as described in section 4.

TABLE I.

	Requirement
Appearance	Clear, and free of suspended matter and sediment.
Color	Not darker than solution containing 0.003 gram potassium dichromate per liter of distilled water.
Copper corrosion	No marked blackening or iridescence
Distillation range	45.5° to 47.5° C.
Nonvolatile content	10 mg. max. per 100 ml.
Odor	Characteristic of CS_2
Residual odor	None
Specific gravity	1.262 to 1.267 at 20° C./20° C.
Spotting or staining	None
Water	No turbidity

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling. — A representative composite sample shall be taken from each delivery and forwarded to the laboratory for test. The sample shall be not less than 2 quarts nor more than 2 gallons.

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4.2 Tests (see CAUTION NOTICE, 6.1.1.)

4.2.1 *Appearance.*—Pour some of the thoroughly mixed sample into a clean dry test tube. Allow to stand for 10 minutes at room temperature (20° to 30° C.). Examine by transmitted light for clarity, suspended matter, and sediment.

4.2.2 *Color.* — Prepare a standard color solution by dissolving 0.003 gram of potassium dichromate in a liter of distilled water. Compare the color of the sample with that of the standard in 50-ml. Nessler tubes at room temperature (20° to 30° C.).

4.2.3 *Copper corrosion.*—The sample shall be tested by method 530.2 of Federal Specification VV–L–791, except that the time of exposure shall be 30 minutes. Examine for compliance with the requirements of table I.

4.2.4 *Distillation range.* — Determine in accordance with method 100.1 of Federal Specification VV–L–791, except that the thermometer used shall conform to the requirements of method 950.1 for type 15C–39, low softening point. The distillation flask shall be heated by a water bath maintained at about 60° C.

4.2.5 *Nonvolatile content.*—Place 100 ml. of the sample in a tared porcelain evaporating dish. Evaporate over a steam bath to dryness. Determine the weight of the residue in mg. per 100 ml.

4.2.6 *Residual odor.*—Dip uniform strips of No. 40 Whatman-type filter paper into the sample. Remove and allow to dry for 2 hours in a well-ventilated room at 20° to 30° C. Examine for residual odor.

4.2.7 *Specific gravity.* — Determine at 20° C. by the pycnometer method as given in method 40.2 of Federal Specification VV–L–791. Calculate specific gravity to water at 20° C.

4.2.8 *Spot test.*—Support a piece of No. 40 Whatman-type filter paper so that the center is free of contact. Place 5 drops of the sample on the center of the filter paper and allow to evaporate in a well-ventilated room at 20° to 30° C. out of direct sunlight. Examine for compliance with table I.

4.2.9 *Water.* — Place 5 ml. of the thoroughly mixed sample in a clean, dry, glass-stoppered, graduated cylinder. Add 90 ml. of 60° API gasoline, in 10-ml. portions, shaking thoroughly after each addition. Examine for turbidity.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. — Carbon disulfide shall be packaged and packed in accordance with Interstate Commerce Commission Regulations for this material.

5.2 Marking and labeling.

5.2.1 *Issue packages.*—Each issue package shall be marked with the name of the material, quantity, and name or trade-mark of the contractor. In addition, each package shall be labeled with the following notice:

"CAUTION — EXTREMELY FLAMMABLE AND POISONOUS SOLVENT. Keep away from heat, fire, flame and lights. Avoid breathing of vapor or contact with skin. Store or use only in well-ventilated place. Keep covered when not in use."

5.2.2 *Shipping containers.*—Shipping containers shall be labeled in accordance with Interstate Commerce Commission Regulations. In addition to the caution label, each container shall be marked with the name of the material, the quantity, name of the contractor, and the contract or order number.

6. NOTES

6.1 Intended use.—Carbon disulfide, technical grade, is intended for use as a solvent

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in industrial applications, and for other technical uses.

6.1.1 *CAUTION NOTICE.* — CARBON DISULFIDE IS HIGHLY VOLATILE, HIGHLY FLAMMABLE AND EXTREMELY POISONOUS. DO NOT USE INDOORS EXCEPT UNDER FORCED VENTILATION. KEEP AWAY FROM HEAT, FIRE OR FLAME. AVOID BREATHING OF VAPOR OR CONTACT WITH SKIN.

When conditions permit, the fire hazard connected with carbon disulfide may be reduced by mixing with carbon tetrachloride.

6.2 **Ordering data.** — Purchasers should specify the size and type of package and shipping container.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications, and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments:

7.2 **Army and Air Force.**—For Army and Air Force purchases, Military Standard MIL–STD–129, Marking of Shipments, shall apply.

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.3 **Navy.**—For Navy purchases, the following specifications, other publications, and special requirements shall apply:

7.3.1 *Applicable specifications and other publications.*

Military Specifications:

JAN–P–110—Packaging and Packing for Overseas Shipment — Drums, Metal, 55-gallon (for Other Than Petroleum Products).

JAN–P–124—Packaging and Packing for Overseas Shipment — Containers (Cans, Pails, and Drums), Metal (for Other Than Subsistence Items).

Navy Department Specification:

General Specification for Inspection of Material.

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.3.2 *Inspection procedures.* — For Naval purchases, the general inspection procedures shall be in accordance with Navy Department General Specifications for Inspection of Material.

7.3.3 *Packing for domestic or overseas shipment.* — Carbon disulfide shall be

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furnished in 5-gallon cans conforming to type V of Military Specification JAN–P–124 or in 55-gallon drums conforming to Military Specification JAN–P–110.

7.3.4 *Marking.*—In addition to any special marking required by the contract or order, shipping containers for the Navy shall be marked in accordance with Military Standard MIL–STD–129.

NAVY INTEREST: *MC* Or S Y

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

For sale by the Superintendent of Documents, Washington 25, D. C. Price cents.

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